

NASA WALLOPS, VA, USA

WMO: 724020

Lat: 37.937N

Lon: 75.466W

Elev: 46

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 93739

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.3	21.4	0.7	5.7	20.0	5.5	7.3	24.4	27.4	39.2	24.7	38.9	9.0	320	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	91.2	76.7	88.6	75.9	86.3	75.0	79.6	86.7	78.3	84.8	77.3	83.4	9.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.6	143.9	83.1	76.5	138.2	82.0	75.4	133.2	81.0	43.0	86.5	41.7	84.8	40.6	83.4	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.8	18.8	10.7	95.3	5.4	2.7	6.9	97.2	3.7	98.8	0.7	100.3	-3.2	102.3
			9.1	81.7	4.9	1.5	5.5	82.7	2.7	83.6	-0.1	84.4	-3.7	85.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	37.7	39.8	45.8	55.2	64.3	73.5	78.0	76.7	71.0	60.5	49.6	42.4
	DBStd	16.07	10.02	8.67	8.53	7.66	7.23	6.16	4.80	4.49	5.81	7.84	8.07	8.79
	HDD50	1299	398	302	187	31	0	0	0	0	0	7	104	270
	HDD65	3934	848	706	595	306	104	9	0	0	16	184	463	703
	CDD50	4214	16	16	57	189	444	705	867	829	631	334	93	33
	CDD65	1374	0	0	1	14	83	264	402	364	197	46	2	1
	CDH74	10508	0	1	9	95	486	1953	3648	2992	1134	186	2	2
	CDH80	3085	0	0	0	17	109	598	1259	857	213	32	0	0
	WSAvg	9.1	9.3	9.7	10.6	10.7	9.5	8.9	8.2	7.9	8.8	8.6	8.5	8.5
	PrecAvg	39.8	3.3	3.0	4.1	2.9	3.3	3.0	3.9	4.0	3.3	2.8	2.6	3.4
(15) Precipitation	PrecMax	58.7	7.3	6.1	6.9	5.0	8.1	7.6	12.1	13.1	9.8	6.9	7.4	8.7
	PrecMin	29.2	0.9	0.7	0.3	0.4	0.4	0.9	0.3	0.6	0.7	0.0	0.0	0.5
	PrecStd	7.6	1.6	1.3	1.6	1.1	2.0	1.6	2.3	2.8	2.2	1.7	1.7	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	68.7	74.5	82.3	87.3	93.1	95.9	93.0	88.9	84.1	72.3	70.1
		MCWB	59.6	59.9	61.5	65.6	71.9	75.9	77.5	78.3	76.6	72.1	64.3	62.4
	2%	DB	60.9	61.9	67.7	76.3	83.1	89.7	91.9	89.6	85.0	78.6	68.2	63.6
		MCWB	55.1	54.4	57.2	64.1	71.0	75.1	77.3	76.9	74.5	69.8	62.1	58.8
	5%	DB	56.6	56.6	62.7	71.4	79.4	86.3	89.0	87.0	82.2	75.3	65.1	59.5
		MCWB	52.4	51.0	54.8	61.6	68.9	73.9	76.5	75.8	73.4	67.9	59.8	56.2
	10%	DB	52.8	52.8	58.4	67.0	75.9	83.1	86.4	84.7	79.9	72.3	62.3	56.1
		MCWB	49.0	48.0	52.4	58.9	67.2	72.7	75.6	75.0	72.2	66.1	57.9	52.9
	0.4%	WB	60.3	61.8	63.8	69.9	75.1	78.9	81.6	80.9	79.4	74.9	67.6	64.7
		MCDB	65.4	67.2	71.5	77.9	83.1	86.8	90.6	87.9	85.1	79.7	70.1	68.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	56.6	56.1	59.5	66.0	72.8	77.3	79.9	79.4	77.3	72.2	64.1	60.3
		MCDB	59.8	59.4	65.6	73.4	80.0	85.5	87.6	86.1	81.7	76.9	66.5	62.7
	5%	WB	53.3	52.3	56.8	63.2	70.7	76.1	78.4	78.1	75.9	69.9	61.6	56.9
		MCDB	55.6	55.3	60.9	69.4	77.2	83.3	85.4	84.0	80.1	73.9	64.1	58.9
	10%	WB	49.7	48.8	53.5	60.6	68.8	74.8	77.4	77.0	74.4	67.8	58.8	53.3
		MCDB	52.2	52.2	57.3	65.9	74.7	81.0	83.9	82.4	78.4	71.0	61.6	55.7
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	15.8	15.9	16.6	15.6	14.5	14.2	14.0	14.4	15.9	16.3	15.8
		MCDBR	19.3	21.1	21.6	22.1	19.7	17.7	16.7	15.5	15.2	16.6	17.5	18.7
		MCWBR	15.9	16.3	14.6	11.7	10.0	7.7	7.1	6.8	7.9	9.8	13.3	15.7
	5% WB	MCDBR	18.3	19.5	19.4	19.3	17.7	15.2	14.5	13.5	12.5	13.6	15.3	17.4
		MCWBR	16.6	17.1	15.1	12.1	10.2	7.5	7.0	6.6	7.2	9.2	13.6	16.1
(40) Clear-Sky Solar Irradiance	taub		0.317	0.323	0.352	0.393	0.439	0.481	0.508	0.502	0.409	0.361	0.329	0.318
	taud		2.510	2.488	2.441	2.340	2.245	2.177	2.117	2.120	2.367	2.497	2.534	2.540
	Ebn at Noon		275	288	289	281	269	257	248	247	266	271	269	267
	Edh at Noon		25	29	33	39	44	47	49	48	36	29	24	23
(44) All-Sky Solar Radiation	RadAvg		699	960	1299	1658	1873	2041	1969	1773	1445	1103	793	617
	RadStd		60	97	105	114	186	144	94	108	129	109	67	54

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.15	N/A	N/A	N/A	N/A	N/A	N/A	-218	+430	+239
(47) Regional (68 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+274	+173

Nomenclature: See separate page